Chapter 7. Inference in Multiple Regression I

Gauss–Markov, confidence intervals, general linear hypotheses, residuals

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Throughout, $\mathbf{y} = X\beta + \epsilon$ with X of size $n \times (p+1)$ and full column rank. We distinguish two sets of assumptions:

- **(A)** $\epsilon \sim N_n(\mathbf{0}, \sigma^2 I_n)$, needed for exact t and F inference;
- **(B)** $E\epsilon = \mathbf{0}$ and $\operatorname{Var}\epsilon = \sigma^2 I_n$ only, enough for the optimality of least squares.

7.1 The Gauss–Markov theorem

Theorem 7.1 (Gauss–Markov)

Under (B), among all linear unbiased estimators $\tilde\beta = A\mathbf{y}$ of β, the least squares estimator $\hat\beta = (X'X)^{-1}X'\mathbf{y}$ has the smallest variance: $\operatorname{Var}\tilde\beta - \operatorname{Var}\hat\beta$ is nonnegative definite. Equivalently, for every $\mathbf{c}$, $\mathbf{c}'\hat\beta$ has the smallest variance among linear unbiased estimators of $\mathbf{c}'\beta$.

Proof. $E(A\mathbf{y}) = AX\beta = \beta$ for all β forces $AX = I$. Write $A = (X'X)^{-1}X' + D$; then $DX = AX - I = 0$. Hence

$$\operatorname{Var}(A\mathbf{y}) = \sigma^2 AA' = \sigma^2\{(X'X)^{-1} + (X'X)^{-1}X'D' + DX(X'X)^{-1} + DD'\} = \sigma^2(X'X)^{-1} + \sigma^2 DD',$$

and DD' is nonnegative definite, with $\mathbf{c}'DD'\mathbf{c} = \|D'\mathbf{c}\|^2$. Equality for all $\mathbf{c}$ holds iff $D = 0$. ■

So $\hat\beta$ is the **best linear unbiased estimator** (BLUE). The theorem needs neither normality nor independence, only uncorrelated errors with a common variance. If $\operatorname{Var}\epsilon = \sigma^2 V$ with $V \neq I$, the BLUE is the generalized least squares estimator (Chapter 12).

7.2 Confidence intervals

Under (A), with $(X'X)^{-1} = (c_{ij})$ and $MSE = SSE/(n-p-1)$ independent of $\hat\beta$ (Theorem 6.4):

Theorem 7.2

$100(1-\alpha)\%$ confidence intervals are:

(a) for a coefficient: $\hat\beta_j \pm t_{\alpha/2}(n-p-1)\sqrt{c_{jj}MSE}$;

(b) for a linear combination $\mathbf{q}'\beta$: $\mathbf{q}'\hat\beta \pm t_{\alpha/2}(n-p-1)\sqrt{\mathbf{q}'(X'X)^{-1}\mathbf{q}\,MSE}$;

(c) for the mean response at $\mathbf{x}$: $\mathbf{x}'\hat\beta \pm t_{\alpha/2}(n-p-1)\sqrt{\mathbf{x}'(X'X)^{-1}\mathbf{x}\,MSE}$;

(d) a prediction interval for a new observation at $\mathbf{x}$: $\mathbf{x}'\hat\beta \pm t_{\alpha/2}(n-p-1)\sqrt{\{1+\mathbf{x}'(X'X)^{-1}\mathbf{x}\}MSE}$.

With σ^2 known, replace MSE by σ^2 and $t_{\alpha/2}$ by $z_{\alpha/2}$.

Proof. (b) contains (a) and (c) as special cases. $\mathbf{q}'\hat\beta \sim N(\mathbf{q}'\beta, \sigma^2\mathbf{q}'(X'X)^{-1}\mathbf{q})$ is independent of $(n-p-1)MSE/\sigma^2 \sim \chi^2(n-p-1)$, so the studentized ratio is $t(n-p-1)$.

(d) adds the variance of the new error. ■

7.3 The general linear hypothesis

Many hypotheses are linear restrictions on β:

$$H_0 : C\beta = \mathbf{m} \quad \text{vs} \quad H_1 : C\beta \neq \mathbf{m},$$

with C a known $k\times(p+1)$ matrix of rank k and $\mathbf{m}\in\mathbb{R}^k$. Examples, for $E(y)=\beta_0+\beta_1x_1+\cdots+\beta_4x_4$:

- $\beta_1=\beta_2$ and $\beta_3=2\beta_4$: $C=\begin{pmatrix}0&1&-1&0&0\\0&0&0&1&-2\end{pmatrix}$, $\mathbf{m}=\mathbf{0}$;
- $\beta_1=\beta_2=\beta_3=\beta_4$: three rows $(0,1,-1,0,0)$, $(0,0,1,-1,0)$, $(0,0,0,1,-1)$;
- $\beta_j=0$: $C=\mathbf{e}_j'$, the j-th unit row.

There are two equivalent routes: minimize SSE under the restriction, or reparametrize to a reduced model.

7.3.1 Method I: constrained least squares

Theorem 7.3 (Restricted least squares)

The minimizer of $\|\mathbf{y}-X\beta\|^2$ subject to $C\beta=\mathbf{m}$ is

$$\tilde\beta=\hat\beta-(X'X)^{-1}C'\{C(X'X)^{-1}C'\}^{-1}(C\hat\beta-\mathbf{m}),$$

and its residual sum of squares is

$$SSE(\tilde\beta)=SSE+Q,\qquad Q=(C\hat\beta-\mathbf{m})'\{C(X'X)^{-1}C'\}^{-1}(C\hat\beta-\mathbf{m}).$$

Proof. With a Lagrange multiplier vector $\lambda\in\mathbb{R}^k$, minimize $L=(\mathbf{y}-X\beta)'(\mathbf{y}-X\beta)+2\lambda'(C\beta-\mathbf{m})$. Setting derivatives to zero:

$$X'X\tilde\beta+C'\lambda=X'\mathbf{y},\qquad C\tilde\beta=\mathbf{m}.$$

The first gives $\tilde\beta=\hat\beta-(X'X)^{-1}C'\lambda$, and substituting into the second, $C\hat\beta-C(X'X)^{-1}C'\lambda=\mathbf{m}$, so $\lambda=\{C(X'X)^{-1}C'\}^{-1}(C\hat\beta-\mathbf{m})$. ($C(X'X)^{-1}C'$ is invertible since C has full row rank.) That this stationary point is the constrained minimum follows from the decomposition below. For any β,

$$\|\mathbf{y}-X\beta\|^2=\|\mathbf{y}-X\hat\beta\|^2+(\hat\beta-\beta)'X'X(\hat\beta-\beta),$$

since $X'(\mathbf{y}-X\hat\beta)=\mathbf{0}$. With $\beta=\tilde\beta$ and $\hat\beta-\tilde\beta=(X'X)^{-1}C'\lambda$,

$$(\hat\beta-\tilde\beta)'X'X(\hat\beta-\tilde\beta)=\lambda'C(X'X)^{-1}C'\lambda=(C\hat\beta-\mathbf{m})'\{C(X'X)^{-1}C'\}^{-1}(C\hat\beta-\mathbf{m})=Q.$$

For any feasible β, $(\hat\beta-\beta)'X'X(\hat\beta-\beta)\geq Q$ (it is the squared distance, in the $X'X$ metric, from $\hat\beta$ to the affine set $\{C\beta=\mathbf{m}\}$, attained at $\tilde\beta$), so $\tilde\beta$ is optimal. ■

Theorem 7.4 (F test of the general linear hypothesis)

Under (A),

$$\frac{Q}{\sigma^2}\sim\chi^2(k,\lambda),\qquad \lambda=\frac{(C\beta-\mathbf{m})'\{C(X'X)^{-1}C'\}^{-1}(C\beta-\mathbf{m})}{2\sigma^2},$$

Q is independent of SSE, and

$$F_0=\frac{Q/k}{SSE/(n-p-1)}=\frac{Q}{k\cdot MSE}\sim F(k,n-p-1,\lambda).$$

Under H_0, $\lambda=0$, and the test rejects when $F_0>F_\alpha(k,n-p-1)$.

Proof. $C\hat\beta-\mathbf{m}\sim N(C\beta-\mathbf{m},\sigma^2C(X'X)^{-1}C')$, and Q is the corresponding quadratic form with matrix $W=\{C(X'X)^{-1}C'\}^{-1}$. Since $W\cdot\sigma^2C(X'X)^{-1}C'/\sigma^2=I_k$ is idempotent, Theorem 5.5 gives the noncentral chi-square with k degrees of freedom. Q is a function of $\hat\beta$, which is independent of SSE (Theorem 6.4). ■

7.3.2 Method II: full and reduced models

Substitute the restriction into the model to get a **reduced model** with $p+1-k$ free parameters; fit it and compute

$$F_0 = \frac{\{SSE(R) - SSE(F)\}/k}{SSE(F)/(n-p-1)},$$

where $SSE(F) = SSE$ is from the full model, with $df_F = n-p-1$ and $df_R = n-p-1+k$. If the restriction involves a nonzero $\mathbf{m}$, move the known part to the left-hand side (an offset). Since $SSE(R)$ is exactly the constrained minimum $SSE(\tilde{\beta})$, Theorem 7.3 gives $SSE(R) - SSE(F) = Q$: *the two methods give the same statistic.*

For a model without intercept, $\mathbf{y} = X\beta + \epsilon$ with X of size $n \times p$, the same holds with $n-p$ in place of $n-p-1$.

Example 7.1

A no-intercept model $y = \beta_1 x_1 + \beta_2 x_2 + \epsilon$ is fitted to

x_1	1	2	0	1	3
x_2	0	1	1	2	1
y	3	7	2	5	10

Test $H_0 : \beta_1 = 2\beta_2$. Here

$$X'X = \begin{pmatrix} 15 & 7 \\ 7 & 7 \end{pmatrix}, \qquad X'\mathbf{y} = \begin{pmatrix} 52 \\ 29 \end{pmatrix}, \qquad \hat{\beta} = \frac{1}{56}\begin{pmatrix} 7 & -7 \\ -7 & 15 \end{pmatrix}\begin{pmatrix} 52 \\ 29 \end{pmatrix} = \begin{pmatrix} 2.875 \\ 1.268 \end{pmatrix},$$

$\mathbf{y}'\mathbf{y} = 187$, and $SSE = 187 - \hat{\beta}'X'\mathbf{y} = 0.732$ with 3 degrees of freedom, so $MSE = 0.244$. *Method I*: $C = (1, -2)$, $C\hat{\beta} = 0.339$, $C(X'X)^{-1}C' = \frac{7+28+60}{56} = 1.696$, so $Q = 0.339^2/1.696 = 0.068$ and $F_0 = 0.068/0.244 = 0.28$. *Method II*: under H_0, $y = \beta_2(2x_1 + x_2) + \epsilon$; with $z = 2x_1 + x_2 = (2, 5, 1, 4, 7)$, $\sum z^2 = 95$ and $\sum zy = 133$, so $SSE(R) = 187 - 133^2/95 = 0.800$ and $\{SSE(R) - SSE\}/MSE = 0.068/0.244 = 0.28$. Both give $F_0 = 0.28 < F_{0.05}(1, 3) = 10.13$: H_0 is not rejected.

7.4 Properties of residuals

Theorem 7.5

For the fitted model with intercept:

(a) $\sum_i e_i = 0$, $\sum_i x_{ij} e_i = 0$ for each j, and $\sum_i \hat{y}_i e_i = 0$;

(b) under (A), $\mathbf{e} = (I - H)\mathbf{y} \sim N_n(\mathbf{0}, \sigma^2(I - H))$; in particular $\operatorname{Var} e_i = \sigma^2(1 - h_{ii})$ and $\operatorname{Cov}(e_i, e_j) = -\sigma^2 h_{ij}$.

Proof. (a) is $X'\mathbf{e} = \mathbf{0}$ together with $\hat{\mathbf{y}}'\mathbf{e} = \hat{\beta}'X'\mathbf{e}$.

(b) $E\mathbf{e} = (I - H)X\beta = \mathbf{0}$ and $\operatorname{Var}\mathbf{e} = \sigma^2(I - H)(I - H)' = \sigma^2(I - H)$. ■

So residuals are correlated, with correlation $-h_{ij}/\sqrt{(1 - h_{ii})(1 - h_{jj})}$, and have unequal variances depending on the design. Plots of residuals against $\hat{y}_i$, against each x_{ij}, and against time order are the basic checks:

- a horizontal band: no problem;
- a funnel: nonconstant variance; use weighted least squares or transform y;

- a linear trend against x_j: a computational error, since (a) forces zero correlation;
- a curve: add a quadratic term (or other function) of x_j, or transform.